

Python notes1

Data Types:

1. Integer (int):

Whole numbers without decimals.

Ex: age = 25

```
print(age)
```

2. Boolean (bool):

A data type that can have only one of two values: True or False.

Ex: is_student = True

```
print(is_student)
```

3. String (str):

A sequence of characters used to represent text. Text enclosed in quotes.

name = "John"

```
print(name)
```

4. Float (float):

A number that contains a decimal point.

Ex: price = 19.99

```
print(price)
```

5. Complex:

A complex number is a number that has a real part and an imaginary part.

Variable names:

A variable in python is defined through assignment.

Ex: Age = 20. Age is a variable name.

Rules for variable names:

- **Can contain letters, numbers and underscores (_)**

Ex: student_name = "John"

- **Must start with a letter or underscore.**

Ex: _name = "Sam"

- **Cannot start with a number.**

Ex: 2name = "Sam" # Error

Type Casting:

Type casting is the process of converting a value from one data type to another.

i) Implicit Type Casting:

Python automatically converts one data type to another without the programmer's intervention.

```
Ex: x = 10    # int
      y = 2.5  # float
      result = x + y
      print(result)
      print(type(result))
```

Output:

```
12.5
<class 'float'>
```

ii) Explicit Type Casting:

The programmer manually converts a value from one data type to another using functions such as int(), float(), str().

```
Ex: x = "25"
      y = int(x)
      print(y)
      print(type(y))
```

Output:

25

<class 'int'>

Dynamic Typing:

means that the data type of a variable is determined automatically at runtime and you do not need to declare the type of a variable before using it.

Ex: x = 10

```
print(type(x))
```

Output: <class 'int'>

x = "Hello"

```
print(type(x))
```

Output: <class 'str'>

Operators in Python:

An operator is a symbol that performs an operation on one or more values (operands) to produce a result.

Ex: a = 10

b = 5

```
print(a + b)
```

i) Arithmetic Operators: It Perform mathematical operations.

Ex: +, -, *, /, %(Modulus → Remainder), //(Floor division → Quotient).

ii) Comparison Operators: It Compare values and return True or False.

Ex: ==, !=, >, <, >=, <=

iii) Logical Operators: It combines conditions.

Ex: AND, OR, NOT

iv) Assignment Operators: Assigning values to variables.

Ex: =, +=, -=, *=, /=

Input function:

The input() function is used to take data from the user during program execution. It always returns the input as a string (str), even if the user enters a number.

```
Ex: name = input("Enter your name: ")  
    print("Hello", name)
```

.format:

.format() is a string method used to insert values into a string in a structured way.

```
Ex: name = "John"  
    age = 20  
    text = "My name is {} and I am {} years old".format(name, age)  
    print(text)
```

Output: My name is John and I am 20 years old.

- {} are placeholders
- .format() fills them in order

f string:

An f string is a way to insert variables or expressions directly inside a string using curly braces {}. It starts with the letter f before the opening quote.

```
Ex: a = 10  
    b = 5  
    print(f"Sum is {a + b}")
```

Output: Sum is 15.

How it works

- f"..." tells Python this is a formatted string.
- {} inside the string are replaced with variable values.

Branching:

Branching is a programming concept where the program makes decisions and chooses different paths of execution based on conditions.

In Python, branching is done using conditional statements like if, elif and else.